



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	9 FEBRUARY 2022
GRADE	11	MARKS	50
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	1 HOUR
CLASS			

MARKING GUIDELINES

Question 1:

- 1.1.1. D ✓
1.1.2. B ✓
1.1.3. D ✓
1.1.4. C ✓
1.1.5. B ✓ (5)

- 1.2.1 Both A and B ✓
1.2.2 B only ✓
1.2.3 A only ✓
1.2.4 None ✓ (4)

- 1.3.1. 1- aorta ✓
4- Renal vein ✓
6- Ureter ✓
7- Urethra ✓ (4)

- 1.3.2. It provides protection ✓ from mechanical injury ✓ (2)

- 1.3.3. Heading ✓
2 differences: ✓✓
- 2 contains oxygenated blood; 4 contains deoxygenated blood
- 2 has unfiltered blood; 4 contains purified blood (can extend on this)
Complete table ✓ (4)

- 1.3.4.
A. More water is reabsorbed ✓ as individuals sweat more, therefore lose more water from the body.
The urine is therefore more concentrated/ thicker ✓ (2)

- B. The dots represent the renal corpuscles ✓ in the cortex. (1)

(13)

Question 2:

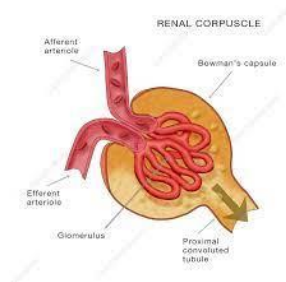
- 2.1.1.
a. D ✓ - has the highest flow rate; contains proteins (any 1 ✓) (2)
b. B ✓ - contains glucose, but no proteins ✓ (2)
c. C ✓ - lowest concentration of sodium ions ✓ (2)
d. A ✓ - lowest flow rate; highest urea concentration; contains ammonium ions (any 1 ✓) (2)

2.1.2. Removal of nitrogenous wastes; regulation of sodium ions/ salt concentration (any 2 ✓✓) (2)

2.1.3. Heading✓

Accuracy and drawing rules ✓

3 x correct labels ✓✓✓



(5)

(15)

2.2.1. Urethra ✓

(1)

2.2.2. Untreated infections can spread ✓ and could cause irreversible/severe damage✓ to the tissue throughout the urinary system. This could result in kidney failure. ✓ Therefore early detection reduces the risk of developing kidney failure. ✓ (any 3)

2.2.3. Cystitis ✓

(1)

2.2.4.

A. The same number of females and males needed to participate in the study.
Each age group needed to contain the same number of males and females
The patients could not be suffering with any other illnesses that could alter the results of the study (any 2✓✓) (2)

B. UTIs are more common amongst young females ✓

(1)

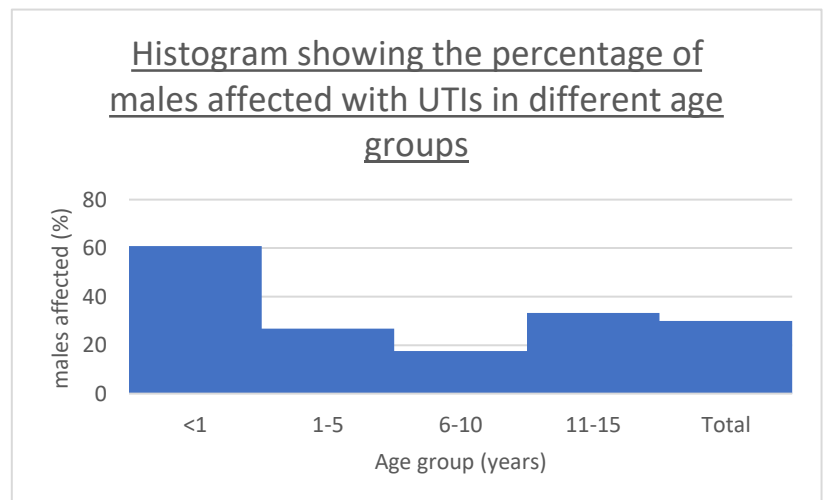
C. Heading✓

Histogram✓

X-axis✓

Y-axis✓

Plotting✓



(5)

(13)

TOTAL: 50 MARKS